Fruity NILA User Manual *version 2025.0.1*

Introduction

Welcome to **Fruity NILA**, a custom hardware integration script for FL Studio that brings deep, intuitive control to your workflow using Native Instruments' Komplete Kontrol including the M-Series, A-Series, and S-Series within FL Studio

Designed to bridge the gap between hardware and software, Fruity NILA lets you take command of FL Studio's key functions—channel rack, mixer, playlist, browser, plugins, and more—directly from your controller. Whether you're mixing, arranging, recording, or performing, this script transforms your controller into a hands-on production powerhouse.

With context-aware control, visual feedback, and powerful shortcut functions, Fruity NILA is the ultimate companion for FL Studio users seeking a hardware-native feel. Crucially, this script operates independently of the Komplete Kontrol application or plugin, ensuring it doesn't disrupt their regular operation.

Special Thanks

Fruity NILA wouldn't be where it is without the incredible help, insight, and encouragement from the community. Hobyst for their outstanding documentation, programs, script NI Host Integration Agent API for FL Studio, and coding assistance.

Note: FL Studio is developed by Image-Line NV. Fruity NILA is an independent project and is not affiliated with or endorsed by Image-Line.

System Requirements

To use Fruity NILA, ensure the following requirements are met:

Software:

- Komplete Kontrol v2.3.0 or higher
- Device Firmware 0.3.9 or higher
- FL Studio 2025 or higher

Hardware:

- **Komplete Kontrol S-Series** Mark II & III (S49/61/88)
- **Komplete Kontrol A-Series** (A25/49/61)
- **Komplete Kontrol M32**

Installation & Setup

To install the *Fruity NILA* script:

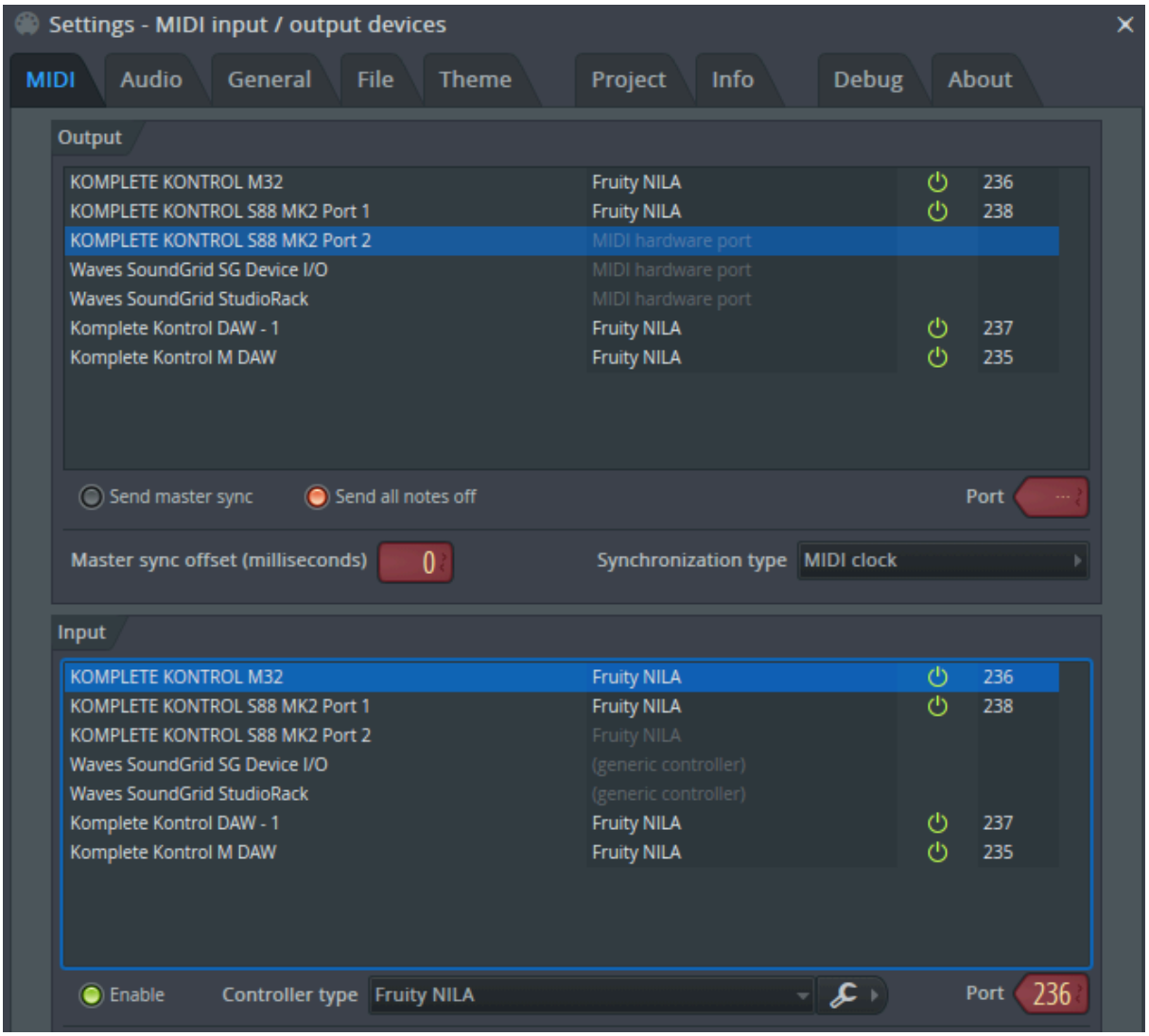
1. Download: github.com/soundwrightpro/FRUITY-NILA/releases/latest. Unzip the downloaded file and locate the FRUITY-NILA folder and open it.

2. Copy the entire ***'Fruity NILA'*** Folder to:

   - **Windows:** Documents\Image-Line\FL Studio\Settings\Hardware
   - **macOS:** ~/Documents/Image-Line/FL Studio/Settings/Hardware

3. Launch **FL Studio**.

4. Navigate to **MIDI Settings**, and set MIDI Tab to setting shown below (M32 & S Series shown, A Series looks similar):
5.



Screenshots of FL Studio are © Image-Line NV and are used for illustrative purposes only.

Tip: Port can be whatever number you want, as long as it's not used by another device.

6. Close the settings. The script will now auto-load when the controller is connected.

7. In the event that auto detection fails, select your control and set 'Controller type' as **Fruity NILA**

Controller Overview

The Fruity NILA script is optimized for Komplete Kontrol keyboards and features dynamic mapping of FL Studio elements to the hardware:

- **Knobs:** Control volume, panning, and plugin parameters.
- **Wheel:** Scroll: Context-aware navigation. In Mixer view, scrolling also attempts to keep the selected track visible by following the viewport.
- **Buttons:** Trigger playback, quantization, undo, plugin pickers, and more.
- **Shift Modifiers:** Access alternate functions using the Shift button.
- **Screen Feedback:** display updates dynamically with track, plugin, and mode info.

Button Controls

Transport

Button	Function	Shift + Button
Play	Starts or pauses the playback. Play LED pulses to tempo while playing.	**Restart**: Returns to start without stopping playback.
Stop	Stop the playback. Stop LED lights on press.	**Clear**: Acts as Escape.
Record (Rec)	Activates recording. REC LED pulses to tempo while active.	**Count-In**: Toggles pre-record countdown.
Loop	Toggles Pattern/Song mode. LED indicates Pattern mode.	–
Metronome (Metro)	Toggles metronome. LED lit when active.	–
Tempo Tap	Functions like FL Studio's tap tempo.	–

Editing & Utility

Button	Function	Shift + Button
Undo	Undoes the last action.	**Redo**: Redoes the last undone action.
Quantize	Triggers quick quantization of selected notes.	**Snap Auto**: Cycles global snap settings.

M (Mute)	–	Mutes selected track (Mixer or Channel Rack).
S (Solo)	–	Solos selected track (Mixer or Channel Rack).

Knob Control

Volume / Pan (S Series only)

Action	Function
Twist	Adjusts volume of the selected track (Mixer or Channel Rack).
Shift + Push Up (Encoder Y-Axis) (S Series only)	Enters **Pan Mode**.
Twist (Pan mode)	All 8 knobs control panning for the corresponding tracks.
Shift + Push Down (Encoder Y-Axis (S Series only)	Exits Pan Mode and returns knobs to volume control.

MIDI Mapping Mode

Action	Function
Shift + Plug-In (A Series & M32) **Plugin** (S Series)	Enters MIDI mode (4 pages of assignable knobs).
Track (A Series & M32)	Returns to Fruity NILA
Mixer (S Series mk II) **DAW** (S Series mk III)	Engages Fruity NILA

8 Buttons Above the Display (S-Series Only)

Komplete Kontrol S-Series keyboards, the **8 buttons above the display** are context-sensitive and perform different actions depending on the current FL Studio window.

In Mixer View

- **Press (mk II)**: Arms the corresponding mixer track for recording.
- **Hold M (Mute) + Press (mk II), Press (mk III)**: Mutes the corresponding track.
- **Hold S (Solo) + Press, Hold Shift + Press (mk III)**: Solos the corresponding track.

These buttons always map to the 8 visible mixer tracks shown on the display.

In Channel Rack

- **Press**: Instantly selects and activates the corresponding channel, making it playable with the keyboard.

Wheel Encoder

Navigation

Action	Function
Single Press	Selects a group (if in Channel Rack).
Double Press	Opens context-specific menus (e.g., group menu in Channel Rack, marker menu in Playlist).
Shift + Press (A Series & M32)	Cycles through UI windows: Mixer → Playlist → Browser → Piano Roll (if open) → Channel Rack.
Shift + Scroll (S Series)	Channel Rack → Piano Roll (if open) → Mixer → Playlist → Browser Counterclockwise reverses direction
Shift + Pan Encoder Scroll (S Series, Pan Mode only)	Cycles through generator plugin windows from the Channel Rack. Clockwise moves to the next generator. Counterclockwise moves to the previous generator. The selected generator window is opened and brought into focus when possible.

Window-Specific Navigation

Channel Rack

- **Scroll / Up/Down:** Browse through channels.
- **Left/Right:** Move through the current pattern.
- **Single Press:** Opens selected wrapper window (plugin window)
- **Double Press:** Opens Channel Group Menu.

Playlist

- **Scroll:** Moves the playhead in bar increments.
- **Left/Right:** Jumps between time markers.
  Double Press: Adds/removes marker at current playhead location.

Mixer

- **Scroll:** Scrolls through mixer tracks.
- **Left/Right:** Jumps by groups of 8 tracks.

- **Down:** Toggles Swap L/R Channels on the selected track.
- **Up:** Toggles Reverse Polarity on the selected track. Note: If a popup menu is open, Up and Down perform normal menu navigation instead.

Browser

- **Scroll / Arrows:** Navigate folders and files. When on playable file right previews file, left closes the folder structure.
- **Double Press:** Opens folder or right-clicks file.

Plugins

- **Scroll/Left/Right:** Scroll through settings
- **Up/Down:** Browse through presets. Note: In Mixer view, Up/Down controls are reassigned to track utilities instead of preset browsing.

Plugin Picker

- **Shift + Double Press** (in any mode): Opens plugin picker.
    - **Left/Right:** Change plugin category.
    - **Scroll/Up/Down:** Scroll through plugins.
    - **Single Press:** Load plugin into Channel Rack.

FLEX Plugin Mode

- **Left/Right:** Switch packs/presets.
- **Up/Down:** Browse presets.
- **Double Press:** Select preset.

Parameter Control

Encoder Navigation

- Automatically adapts to context (Generator vs. Effect plugin).
- Supports scrolling through parameters with page-style banking.
- Displays parameter values and plugin names on the screen.

Preset Control

- In supported plugins:
    - **Encoder UP/DOWN:** Loads previous/next preset.
    - **Automatic fallback:** If not a plugin, scrolls up/down.

Display Feedback

Function	Shift + Button
Default	Current track name, volume, mute/solo state.
Shift Held	Replaces volume with panning.
Knob touched	Shows track name.
Playlist Playing	Displays time in bars/beats or minutes/seconds.
Metro/Loop/Snap Buttons	Shows relevant mode/status.
Parameter	Displays plugin name and parameter values.

Komplete Kontrol Plug-In Support

If Komplete Kontrol is open as a plugin:

- All control (wheel, knobs, M/S, buttons) is redirected to the plugin.
- and controller reflect plugin mode.

Instance Switching

Action	Shift + Button
Shift + Track (Instance)	Toggles between FL Studio and KK plugin control.
Wheel Scroll (while holding Shift)	Scrolls through instances of KK.
Single Press (Wheel)	Select instance.
Shift + Plug-In	Switch back to full FL Studio control.

Additional Features

- **Mod Wheel / Pitch Wheel:** Passed through to plugins that support them.
    - Note: The pitch wheel responds according to the pitch range defined in the channel settings. To customize how far the pitch bends, adjust the Pitch Range knob in the Channel Settings window of the selected instrument.

- **Snap Feedback:** shows current snap resolution when changed.
- **Live Parameter Display:** Active parameter info shown during control.
- **Auto Fallback:** Invalid or unsupported plugins are automatically skipped without error.

User Adjustable Settings

Fruity NILA includes several user-configurable settings that allow you to tailor the script's behavior to match your workflow. These settings can be adjusted by editing the config.py file located inside the script folder:

Location:

macOS:
~~/Documents/Image-Line/FL Studio/Settings/Hardware/FRUITY-NILA/script/NILA_engine/config.py

Windows:
Documents\Image-Line\FL Studio\Settings\Hardware\FRUITY-NILA\script\NILA_engine\config.py

Available Settings:

Setting	Description	Default
double_click_speed	Time in seconds between two clicks to register a double press.	0.5
jog_preview_sound	Play preview sound when using the jog wheel in the Browser.	0
upDown_preview_sound	Play preview sound when using the up/down wheel in the Browser or Playlist.	1
rectChannel	Duration (in ms) to show the rectangle over the selected Channel Rack slot.	2000
rectMixer	Duration (in ms) to show the rectangle over the selected Mixer track.	2000
mixer_increment	Amount of volume or pan change per knob twist in the Mixer. Range: 0.001–1.00.	0.005
channel_increment	Amount of volume or pan change per knob twist in the Channel Rack. Range: 0.005–1.00.	0.01

> **Tip:** After changing settings in config.py, restart FL Studio or reload the script for changes to take effect.

Limitations & Notes

- Buttons like *Scale*, *Arp*, and *Ideas* are exclusive to Komplete Kontrol and not mapped in Fruity NILA.
- Preset navigation only works with supported plugins.
- Plugin Picker is not supported in detached plugin windows.

- FLEX, Komplete Kontrol, and other proprietary plugins have context-sensitive navigation handled internally.
- Single-click actions are slightly delayed to determine if a double-click occurs. Prevents both actions from triggering at the same time.
- Some advanced features such as Pan Mode are only available on S Series keyboards.

Supported Plugins

This includes (but is not limited to): FLEX, FPC, Fruity Parametric EQ 2, Harmor, Sytrus, Harmless, GMS, Fruity Delay Bank, Fruity Reeverb 2, Fruity Limiter, Gross Beat, Serum, Vital, Komplete Kontrol, FabFilter Pro-Q, RC-20 Retro Color, OTT, ShaperBox, Neutron, Ozone, Decapitator, Microtonic, Valhalla Room, EchoBoy, Arturia Pigments, U-he Diva

These plugins are supported as long as they provide valid parameter data and are not explicitly listed as unsupported. The script will correctly display and allow parameter browsing and paging.

Unsupported Plugins

This includes (but is not limited to): Fruity Send, Fruity Stereo Shaper, Fruity Notebook, Fruity Video Player, Fruity Big Clock, Fruity Spectroman, Fruity WaveShaper, Patcher, Control Surface, MiniHost Modular, Fruity Dashboard

These plugins are unsupported due to limitations in how they expose parameters, metadata, or UI integration. The script automatically skips them during navigation to prevent errors or UI issues.

Troubleshooting

Issue: Encoder not responding
Fix: Ensure the correct window is in focus. The encoder is context-sensitive.

Issue: “Plugin not valid” error
Fix: This occurs if an unsupported or unrecognized plugin is selected. Switch to a supported plugin or track.

Issue: Some knobs/buttons not working
Fix: Ensure MIDI mode is set to *Track*. Use **Shift + Plugin** to enter MIDI mode, **Shift + Track** to exit.

Issue: Komplete Kontrol not switching to FL Studio
Fix: Use **Shift + Track** to toggle between Komplete Kontrol and Fruity NILA.

Support & Community

Need help? Want to suggest a feature, report a bug, or chat with other users?

Join the **Fruity NI on Discord** for support, updates, and collaboration:
https://discord.com/invite/GeTTWBV

Glossary of Terms

Term	Definition
FL Studio	A digital audio workstation (DAW) developed by Image-Line.
Channel Rack	A section in FL Studio used to host generators (instruments and samples).
Mixer	Where audio levels, routing, and effects are managed for individual tracks.
Playlist	The arrangement area for patterns, audio clips, and automation.
Browser	The file and plugin browser in FL Studio.
Generator Plugin	A plugin that generates sound, such as FLEX or Sytrus.
Effect Plugin	A plugin used to process audio (e.g., EQs, reverbs, delays).
Komplete Kontrol	A series of MIDI keyboard controllers from Native Instruments.
Display	The screen on supported controllers used for visual feedback.
Jog Wheel	A scroll/rotate wheel used to navigate through different parts of FL Studio.
Preset	A saved configuration of a plugin's parameters.
Track	A mixer or channel slot that holds instruments or effects.
Parameter	A controllable setting of a plugin (e.g., filter cutoff, resonance).
Lead Param	The first plugin parameter shown on screen when browsing via encoder.
Unsupported Plugin	A plugin whose parameters cannot be read or mapped reliably by Fruity NILA.
Shift Modifier	Access alternate functions and modes using the Shift button. On S Series devices, this includes entering and exiting Pan Mode.
Pan Mode	A control mode available on S Series devices where all 8 knobs adjust panning instead of volume. Activated with Shift + Encoder Up and exited with Shift + Encoder Down.
Selected Encoder	The main push encoder used for navigation and control. Supports directional input (left, right, up, down) and press actions, including Shift-modified functions.
Generator Cycling	A navigation function that cycles through generator plugin windows in the Channel Rack using Shift + Pan Encoder Scroll while in Pan Mode (S Series only).

Screen Label Glossary

Short Form	Meaning	Context
S	Sampler	Generator plugin type
P	Plugin / Generator / Hybrid	Generator plugin type
L	Layer Channel	Generator plugin type
AC	Audio Clip	Generator plugin type
Auto	Automation Clip	Generator plugin type
P | Insert: X	Effect Plugin on Mixer Insert X	Mixer plugin type
B:B | 3:01	Beats:Bar + time position display	Playlist beat counter ()
M:S | 1:22	Min:Sec + time position display	Playlist time display ()
Volume	Channel Volume	Displayed on knob 0
Mix Level	Effect Plugin Wet/Dry Mix	Displayed on main knob

Browser File Types

When browsing through the FL Studio Browser, Fruity NILA can detect different types of files based on the labels shown. These file types are listed below:

File Type	Description
.FLP	FL Studio project file
.ZIP	Zipped FL Studio project
FL Preset	FL Studio instrument or effect preset
.WAV	Uncompressed audio file
.MP3	Compressed audio file
.OGG	Ogg Vorbis audio file
.FLAC	Lossless audio file
.AIFF	Apple audio file format
.WV	WavPack audio file
.SPEECH	Text-to-speech export file
.SF2	SoundFont instrument file
.MIDI	Standard MIDI sequence file
FLEX PACK	Expansion pack for FLEX plugin

.MP4	Video file
.FNV	FL Studio video file format
.TXT	Text file
.INI	Configuration/settings file
.FSC	FL Studio score file (melody or pattern)
IMAGE File	Image formats such as PNG or JPEG
FL Proj.	FL Studio project folder

Disclaimer

The information contained in this document is subject to change without notice and does not represent a commitment on the part of the author. The *Fruity NILA* Python script and its associated documentation are licensed under the Creative Commons Attribution-NonCommercial-ShareAlike 4.0 International License (CC BY-NC-SA 4.0).

You should have received a copy of the license along with this work. If not, see http://creativecommons.org/licenses/by-nc-sa/4.0/.

Under the following terms:

- Attribution — You must give appropriate credit, provide a link to the license, and indicate if changes were made. You may do so in any reasonable manner, but not in any way that suggests the licensor endorses you or your use.
- NonCommercial — You may not use the material for commercial purposes.
- ShareAlike — If you remix, transform, or build upon the material, you must distribute your contributions under the same license as the original.
- No additional restrictions — You may not apply legal terms or technological measures that legally restrict others from doing anything the license permits.

"Fruity NILA" and all related names, features, and logos are trademarks of their respective owners.
FL Studio and related product names are registered trademarks of Image-Line NV.. Native Instruments, Komplete, and related product names are trademarks or registered trademarks of Native Instruments GmbH in Germany and other countries. Mac and macOS are registered trademarks of Apple Inc., registered in the U.S. and other countries. Windows is a registered trademark of Microsoft Corporation in the United States and other countries.

All other trademarks are the property of their respective owners. References to third-party trademarks do not imply endorsement, affiliation, or sponsorship by their respective holders.

Limitation of Liability

The *Fruity NILA* script and accompanying documentation are provided "as is" without warranty of any kind, either express or implied, including, but not limited to, the implied warranties of merchantability, fitness for a particular purpose, or non-infringement.

Use of this software is entirely at your own risk. The author makes no representations or warranties regarding the functionality, compatibility, or safety of the *Fruity NILA* script when used with any specific hardware or software configuration. In particular, there is no guarantee that the script will function without causing unintended behavior, including but not limited to performance issues, communication failures, data loss, or physical malfunction of MIDI controllers or other connected equipment.

By using this software, you acknowledge and agree that the author shall not be held liable for any damages, direct or indirect, incidental, special, exemplary, or consequential, arising out of the use or inability to use the script, even if advised of the possibility of such damages.

You assume full responsibility for testing and deploying the *Fruity NILA* script in your own environment, including safeguarding your hardware and data.